

Hydrogen Production via Water Gas Shift Reaction (WGSR): Aspen HYSYS Simulation

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Background & Description:

This project focuses on the simulation of the Water Gas Shift Reaction (WGSR) in order to enhance hydrogen production from synthesis gas. WGSR is a key downstream process in hydrogen production routes such as Steam Methane Reforming (SMR), where the reformer outlet gas still contains significant amounts of carbon monoxide (CO) and unreacted steam.

The conversion of CO through WGSR not only increases the hydrogen yield but also reduces the concentration of toxic carbon monoxide. Industrially, this reaction is carried out in two successive stages: a High Temperature Shift (HTS) reactor followed by a Low Temperature Shift (LTS) reactor. This configuration allows high overall CO conversion while maintaining favorable reaction kinetics and thermodynamic equilibrium.

Process Simulation in Aspen HYSYS:

The WGSR section of the process was simulated using Aspen HYSYS, assuming that the feed stream corresponds to the outlet of a Steam Methane Reformer. The feed gas mainly consists of hydrogen, carbon monoxide, carbon dioxide, steam, methane, and nitrogen.

Thermodynamic Model

- Peng–Robinson equation of state, selected for its suitability in high-temperature gas-phase systems containing light gases.

Reactor Configuration

Two Conversion Reactors were used to represent the WGSR stages:

1. High Temperature Shift (HTS) Reactor
 - Temperature range: 500–700 K
 - Main reaction:
 $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$
 - Approximately 35% of the residual CO is converted in this stage.
2. Low Temperature Shift (LTS) Reactor
 - Feed gas cooled to 300–400 K before entering the reactor
 - Same reaction mechanism as HTS
 - Final conversion of CO to nearly zero concentration

Downstream Separation

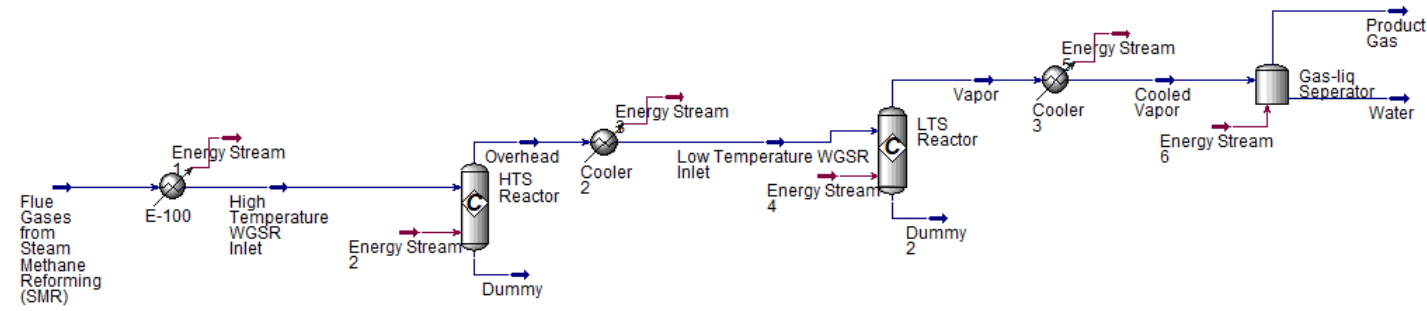
- The reactor outlet stream is cooled and sent to a gas–liquid separator.
- Condensed water is removed, and the dry gas stream is assumed to be further purified using Pressure Swing Adsorption (PSA) to obtain hydrogen-rich product gas.

Conclusion

The Aspen HYSYS simulation confirms the effectiveness of the **Water Gas Shift Reaction** as a post-reforming process for hydrogen enrichment and CO removal. The two-stage HTS–LTS configuration provides high CO conversion while maintaining realistic industrial operating conditions.

This work demonstrates that Aspen HYSYS can successfully model WGSR-based hydrogen purification systems and provides results consistent with reported industrial and academic data.

Flowsheet



Results:

Master Property Tableau								
Object	Flue Gases from SMR	Overhead	Dummy	Low Temperature WGSR	Dummy 2	Vapor	Water	Product Gas
T (K)	617.1	698.1	698.1	471.1	473.1	473.1	299.6	299.6
P (kPa)	405.3	405.3	405.3	405.3	405.3	405.3	405.3	405.3
Molar Flow (kgmole/h)	16.95	16.95	<0	16.95	<0	16.95	2.684	14.27
Phase Comp Molar Flow (Overall-CO) (kgmole/h)	1.2996	1.2996	<0	0.0000	<0	0.0000	0.0000	0.0000
Phase Comp Molar Flow (Overall-CO2)	0.9432	0.9432	<0	2.2428	<0	2.2428	0.0000	2.2420
Phase Comp Molar Flow (Overall-H2O)	4.1076	4.1076	<0	2.8080	<0	2.8080	2.6835	0.1246
Phase Comp Molar Flow (Overall-Hydrogen)	5.3963	5.3963	<0	6.6960	<0	6.6960	0.0000	6.6960

Phase Comp Molar Flow (Overall- Methane)	0.2736	0.2736	<0	0.2736	<0	0.2736	0.0000	0.2736
Phase Comp Molar Flow (Overall- Nitrogen)	4.8060	4.8060	<0	4.8060	<0	4.8060	0.0000	4.8060
Phase Comp Molar Flow (Overall- Oxygen)	0.1260	0.1260	<0	0.1260	<0	0.1260	0.0000	0.1260

Nb : Insignificant quantities (<0) of components in the product streams can be considered negligible, effectively approximating them to zero without impacting the overall process results or conclusions (see flowsheet).

References: Ibañez, J. F. C., Ergueta, P. L. A., Bilbao, M. B. G., & Cagigal, V. L. B. (2011). *U.S. Patent No. 8,075,870*. Washington, DC: U.S. Patent and Trademark Office.